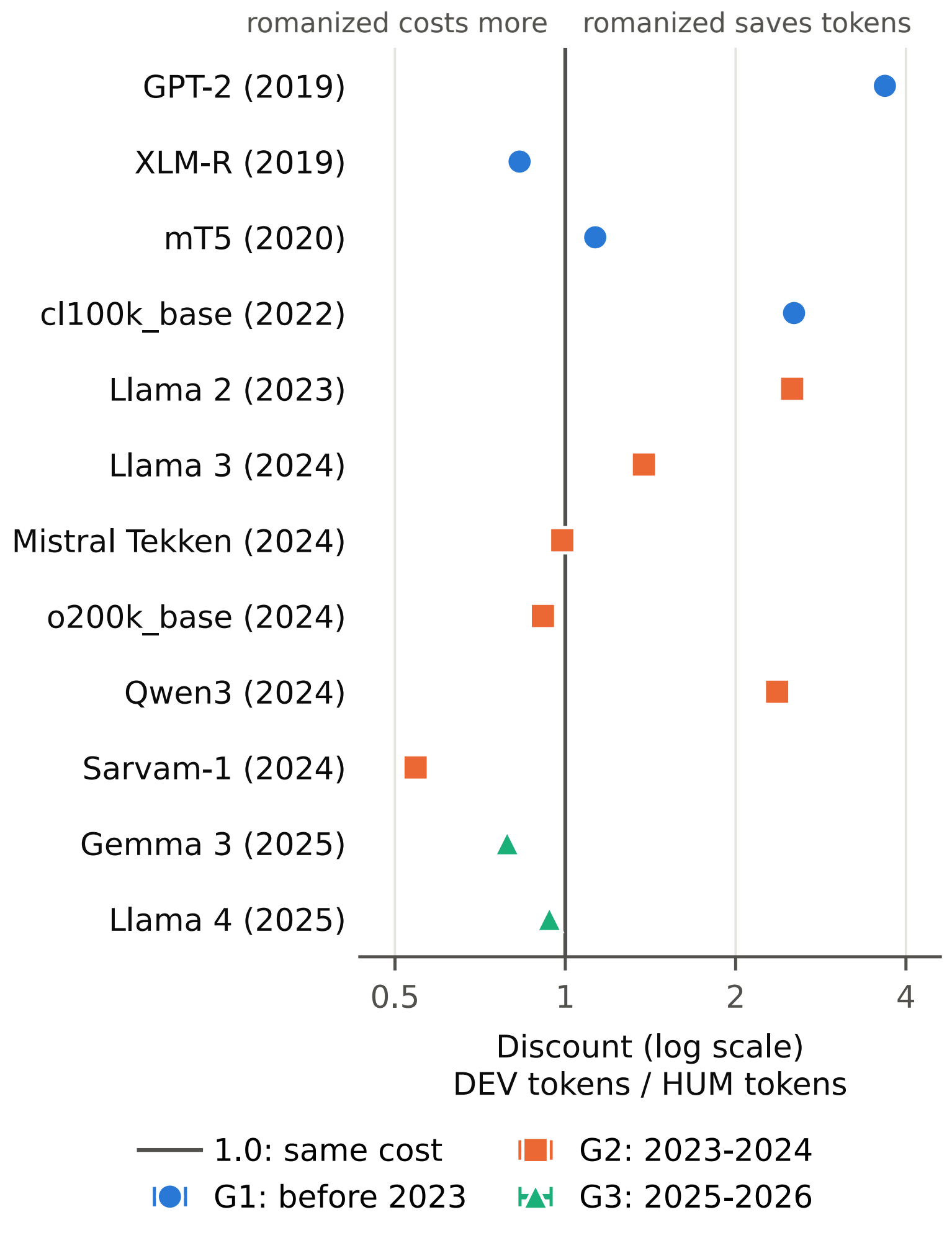
romanized costs more
romanized saves tokens
GPT-2 (2019)
XLM-R (2019)
mT5 (2020)
cl100k_base (2022)
Llama 2 (2023)
Llama 3 (2024)
Mistral Tekken (2024)
o200k_base (2024)
Qwen3 (2024)
Sarvam-1 (2024)
Gemma 3 (2025)
Llama 4 (2025)
0.5
1
2
4
Discount (log scale)
DEV tokens / HUM tokens
1.0: same cost
G1: before 2023
G2: 2023-2024
G3: 2025-2026